

ALISINA BAYATI

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Citizenship: United States | Iran

RESEARCH PROFILE

Research Focus: Developing **Maximum Entropy Principle (MEP)** frameworks based on deterministic annealing (DA), combined with **control-theoretic tools**, to address NP-hard and nonconvex combinatorial optimization problems with applications to routing, capacitated facility location, constrained sparse linear regression, orthogonal NMF, and safety-constrained dynamic coverage.

EDUCATION

University of Illinois at Urbana–Champaign <i>Ph.D. Mechanical Science and Engineering</i> Committee: Prof. Jeff Shamma, Prof. Prashant Mehta, Prof. Yingying Li	2020 – 2026 (Exp.) Advisor: Prof. Srinivasa Salapaka
University of Illinois at Urbana–Champaign <i>M.S. Mathematics</i>	2023 – 2025 GPA: 3.90/4.00
Sharif University of Technology <i>B.S. Mechanical Engineering</i>	2015 – 2019 GPA: 3.89/4.00 (18.10/20), Advisor: Prof. Hamed Moradi

PUBLICATIONS

Peer-Reviewed

1. **A. Bayati**[†], D. Tiwary[†], S. Salapaka. “A Control Barrier Function Approach to Constrained Resource Allocation Problems in a Maximum Entropy Principle Framework.” *IEEE Conference on Decision and Control (CDC)*, 2025. 📄 [Paper](#) 🔄 [Code](#)
2. S. Basiri, **A. Bayati**, S. Salapaka. “Orthogonal Non-negative Matrix Factorization with Sparsity Constraints.” *IEEE Conference on Decision and Control (CDC)*, 2025. 📄 [Paper](#) 🔄 [Code](#)
3. **A. Bayati**, A. Srivastava, V. Mundada, S. Salapaka, H. Feng, A. Malvandi. “Energy Efficiency in Industrial Drying: A Hybrid Ultrasonic System with a Novel Dynamic Optimization Framework.” *Energy Conversion & Management*, 2025. 📄 [Paper](#)
4. A. Srivastava[†], **A. Bayati**[†]. “Sparse Linear Regression with Constraints: A Flexible Entropy-Based Framework.” *European Control Conference (ECC)*, 2024. 📄 [Paper](#)
5. **A. Bayati**, A. Srivastava, A. Malvandi, H. Feng, S. Salapaka. “Towards Efficient Modularity in Industrial Drying: A Combinatorial Optimization Viewpoint.” *American Control Conference (ACC)*, 2023. 📄 [Paper](#)

Working Papers

6. **A. Bayati**, A. Srivastava, S. Salapaka. “Adaptive Resource Allocation under Demand Distribution Shifts: A Maximum Entropy Framework with Safety and Mobility Constraints.”
7. **A. Bayati**, A. Srivastava, S. Salapaka. “Capacitated Facility Location and Routing Optimization: A Maximum Entropy Principle and Control-Theoretic Approach.”

RESEARCH EXPERIENCE

Graduate Research Assistant, SENSIC Lab, University of Illinois Urbana–Champaign	2020 – Present
Control-Theoretic Approaches to Combinatorial Optimization	2024 – Present
– Designed a dynamic feedback strategy for constrained optimization that provably drives decision variables toward KKT stationarity with guarantees on convergence and feasibility. – Applied to capacitated facility location problems, achieving up to $\sim 4\times$ and $\sim 74\times$ speedups over Safe Gradient Flow (Cortés et al.) and SciPy’s SLSQP. 📄 Paper 🔄 Code – Applied the framework to orthogonal non-negative matrix factorization with feature sparsity. 📄 Paper 🔄 Code – Proved almost-sure avoidance of strict saddle points under random initialization. – Ongoing work extends these frameworks to non-stationary environments with shifting demand distributions, aiming to track time-varying optimizers with high sample efficiency.	
Mixed-Integer Control Frameworks for Hybrid Actuator Systems	2020 – Present
– Developed a mixed-integer multi-stage optimization framework for ultrasonic/convective drying, reducing energy consumption by 13% relative to a static single-stage baseline while satisfying product quality constraints (moisture and color). 📄 Journal Paper 📄 ACC Paper – Extending the framework to smart drying systems involving multiple drying technologies, jointly optimizing both the drying sequence and operating conditions. – Integrating mesh-based graph neural networks trained offline to replace CFD/statistical models for fast differentiable inference.	
Routing Optimization in Residential Wi-Fi Mesh Networks	2021 – 2023

- Built a real-time digital twin for Wi-Fi mesh networks using permutation-invariant neural networks to solve large-scale placement and routing problems.
- Achieved network latency in the **lowest 3rd percentile** across 50,000 uniformly sampled brute-force configurations.
- Project funded by *Foxconn Interconnect Technology (FIT)*.

Undergraduate Researcher, Sharif University of Technology

2018 – 2019

- Designed optimal PID controllers for nonlinear cancer drug-delivery models in MATLAB/Simulink to minimize side effects while achieving target tumor reduction.

Notable Course Projects

- **Statistical Reinforcement Learning (CS 542)**, UIUC Fall 2024
 - Studied adaptive control in LQ systems based on [Abbasi-Yadkori & Szepesvári \(2011\)](#); derived core lemmas and reproduced the $\tilde{O}(\sqrt{T})$ regret bound. [\[Report\]](#)

TEACHING & MENTORSHIP

Graduate Teaching Assistant, University of Illinois Urbana–Champaign

2023 – Present

- **Recognition:** Ranked “Excellent” in campus-wide student evaluations (2024).
- **Graduate Course:** *TAM 541 – Mathematical Methods I*. Led recitation sessions for 50+ graduate students covering complex analysis, linear algebra, and ordinary differential equations.
- **Undergraduate Courses:** *ME 340 – Dynamics of Mechanical Systems*, *TAM 210/211 – Statics & Intro to Dynamics*. Managed laboratory sections and incorporated MATLAB-based simulations to support conceptual understanding.

Research Mentorship, SENSIC Lab (UIUC)

2024 – Present

- Mentoring two undergraduate researchers on routing optimization for residential Wi-Fi mesh networks.

SELECTED GRADUATE COURSEWORK

Mathematics	Machine Learning & Statistics	Control & Systems
Real Analysis	Statistical Reinforcement Learning	Theory of Optimal Control
Functional Analysis	Machine Learning	Analysis of Nonlinear Systems
Abstract Algebra	Computer Vision	Linear Control Systems
Abstract Linear Algebra	Probability & Statistics	Dynamic Programming
Optimization by Vector-Space Methods		Random Processes

HONORS & AWARDS

- **Best Presentation Award**, Optimization & Reinforcement Learning Session, CSL Student Conference 2026
- **IEEE Student Travel Award**, Conference on Decision and Control (CDC) 2025
- **IEEE Student Travel Award**, American Control Conference (ACC) 2023
- **Teachers Ranked as Excellent**, campus-wide student evaluations, University of Illinois Urbana–Champaign 2024
- **Graduate Admission Fellowship (First Year)**, Mechanical Science & Engineering Department, UIUC 2020
- **Top 0.2% National Rank**, Iran Nationwide University Entrance Exam (182,000+ participants) 2015

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, pandas, PyTorch, CVXPY, scikit-learn), MATLAB, C, $\text{\LaTeX}$
Scientific Tools	Jupyter, Simulink, OpenCV
Languages	Persian (Native), English (Fluent), Arabic (Basic)

PROFESSIONAL SERVICE

Reviewer: IEEE Conference on Decision and Control (CDC) 2025; American Control Conference (ACC) 2024; European Control Conference (ECC) 2024; Learning for Dynamics and Control (L4DC) 2025.

REFERENCES

Available upon request.